Band-1 Hadamard Order 7156 — Miyamoto (Paley-I DIRECT H_1788 + C_1790) Miyamoto-only (NOT in SPK); Gap 2052 Still OPEN

H_1788 Paley-I PASS; C_1790 PASS; H_7156 Miyamoto PASS; Paley-I/II/Kronecker FAIL → Miyamoto-only (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7108_Miyamoto`; `2026-09-16_Band1_Hadamard_6884_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_7156_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1788 and 7156 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I DIRECT seed H_{1788} = Paley-I($q = 1787$) and symmetric Paley conference C_{1790} over prime $q = 1789 \equiv 1 \pmod 4$. Classical SPK for 7156: Paley-I **FAIL**s ($q = 7155 = 3^3 \cdot 5 \cdot 53$ composite); Paley-II **FAIL**s ($q = 3577 = 7^2 \cdot 73$ NOT prime power); Kronecker **FAIL**s (4×1789). Therefore order **7156 is a Miyamoto-only bonus (NOT in SPK)** — same honesty class as 7108/6836/6788. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7108 Miyamoto-only (NOT in SPK)**. Prior SPK seeds **888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436 / 212 / 424 / 1696** retained; Miyamoto-only bonuses **6676/6484/6404/6388/6196/6788/6836/7108/7156/...** excluded; **1788** folded into ONBOX when verified; **7156** NOT folded.

Abstract

Finite constructions of orders **1788** and **7156**.

- H_{1788} = Paley-I ($q = 1787$) — DIRECT (NOT Double)
- C_{1790} = Paley conference over prime $q = 1789$
- H_{7156} = Miyamoto(H_{1788}, C_{1790})
- Paley-I for 7156: **SKIP** ($q = 7155$ composite)
- Paley-II for 7156: **SKIP** ($q = 3577$ NOT prime power)

SPK: 7156 Miyamoto-only bonus (NOT in SPK). Seed 1788 enters SPK. Prior SPK **888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7156.py`

Results. overall_pass=true; units 108/108; wall 295.275s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 7156 = 4 \cdot 1789$; 1789 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1788} and C_{1790} (Paley conference over prime $q = 1789$).
- Seed: Paley-I $q = 1787$ (prime $\equiv 3 \pmod 4$) → H_{1788} **DIRECT** (NOT Double; Paley-II/Double half FAIL).
- SPK: Paley-I $q = 7155$ **FAIL** (composite $3^3 \cdot 5 \cdot 53$); Paley-II $q = 3577$ **FAIL** ($7^2 \cdot 73$ NOT prime power); Kronecker **FAIL → Miyamoto-only (NOT in SPK)**.
- Prior: 7108 Miyamoto-only shipped; next optional was $q = 1789 \rightarrow n = 7156$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1788 Paley-I	Paley-I q=1787 DIRECT	0	5.0563
C_1790	Paley conf q=1789	0	3.8086
H_7156 Miyamoto	Miyamoto(H_1788 Paley-I DIRECT, C_1790)	0	282.8427
H_7156 Paley-I	SKIP q=7155 composite $3^3 \cdot 5 \cdot 53$	—	—

Object	Method	max|HH^T-nI|	Wall (s)
H_7156 Paley-II	SKIP q=3577=$7^2 \cdot 73$ NOT prime power	—	—
H_7156 Kronecker	SKIP $4 \cdot 1789$ not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **7108** lineage (**888/1776** Paley-I/Double; **860/1720/6884** Paley-I/Double; **1708** Paley-II; **6724** Paley-II; **840/1680**; **6628**; **1668**; **1656**; **1620**; **1608**; **6436**; **212/424/1696**; **NOT** 6676/6484/6404/6388/6196/6788/6836/7108/7156) + verified **1788** (Paley-I DIRECT); **exclude** Miyamoto-only bonuses including **7156**. Known count 1422. Next constructive gap **2052** (OPEN).

Quantity	Value
7156 in SPK	**False** (Miyamoto-only; excluded)
1788 in SPK	**True** (Paley-I q=1787 DIRECT)
888 / 1776 in SPK	**True** (prior 7108 seeds)
860 / 1720 / 6884 in SPK	**True** (prior)
1708 / 840 / 1680 / 6724 in SPK	**True** (prior)
7108 / 6836 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · 7156 not claimed in SPK.

Repro

Lab: `/workspace/experiments/band1/hadamard_7156_miyamoto/`
Artifacts: `H1788_paleyI_pm1.npz`, `C1790_pm1.npz`, `H7156_miyamoto_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.